Uncertainty-Aware Financial Forecasting

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***Abstract*—This paper presents an uncertainty-aware financial forecasting system for the Borsa Istanbul 100 Index (BIST 100) that integrates three methodological pillars: an encoder-only Transformer for multivariate time-series forecasting, Monte Carlo Dropout for epistemic uncertainty quantification, and a hybrid K-Means and Hidden Markov Model pipeline for market regime detection. The system is trained on 4,223 daily observations spanning 2010–2026, combining Turkish equity data, S&P 500 signals, and FRED-MD macroeconomic indicators. Experimental results show that the Transformer achieves a directional accuracy of 51.37% on the 2024–2026 test window, with mean epistemic uncertainty $\bar{\sigma} = 0.0035$. The regime detection module identifies three distinct market states—Bull, Sideways, and Bear—with strong temporal persistence. Although point-forecast metrics do not surpass the naive baseline, consistent with prior literature on daily equity log return prediction, the system delivers meaningful uncertainty estimates and interpretable regime labels that support risk-aware decision-making.**

***Index Terms*—financial forecasting, uncertainty quantification, Monte Carlo Dropout, Transformer, market regime detection, Hidden Markov Model, BIST 100**

I. Introduction

Financial markets are inherently uncertain and volatile. Traditional forecasting models often fail to capture the stochastic nature of financial data and, more critically, do not communicate how confident they are in their own predictions. In practice, a model that acknowledges its uncertainty is more useful for risk management than one that produces overconfident point estimates.

This project addresses these limitations by developing an uncertainty-aware forecasting system for the Borsa Istanbul 100 Index (BIST 100), Turkey's primary equity benchmark. The system combines three components: (i) an encoder-only Transformer that processes multivariate macroeconomic and equity inputs, (ii) Monte Carlo Dropout for principled epistemic uncertainty estimation, and (iii) a hybrid K-Means and Hidden Markov Model pipeline that identifies latent market regimes. Together, these components produce not only a point forecast of the next-day log return but also a confidence measure and a contextual market-state label.

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II. Related Work

A. Transformer Architectures for Financial Time-Series Forecasting

The application of deep learning to financial forecasting has evolved significantly from classical statistical methods toward attention-based architectures. Early approaches relied on autoregressive models such as ARIMA and ensemble methods like Random Forest, which remain competitive baselines due to their interpretability and robustness under strict leakage-controlled evaluation protocols [1]. However, these models struggle to capture nonlinear dependencies and structural breaks characteristic of financial markets. Recurrent architectures, particularly Long Short-Term Memory (LSTM) networks, addressed some of these limitations by modeling sequential dependencies through gated memory mechanisms, yet their performance remains asset-dependent and sensitive to evaluation design [1].

Motivated by the success of self-attention mechanisms in sequence modeling, recent work has explored encoder-only Transformer designs for one-step-ahead return forecasting using multivariate inputs that integrate lagged macroeconomic variables alongside technical indicators [1]. A key limitation of standard single-stream attention is its inability to distinguish between heterogeneous data types such as price signals and macroeconomic features. To address this, Hadizadeh et al. proposed a dual-stream attention architecture consisting of a Price Attention Network (PAN) and a Nonprice Attention Network (NAN), which maintain separate attention pathways for price-based and auxiliary financial inputs respectively [2]. This design provides the structural blueprint for the multivariate integration strategy adopted in the present work, where Borsa Istanbul price data and FRED-MD macroeconomic indicators are treated as complementary input streams.

B. Uncertainty Quantification via Monte Carlo Dropout

Standard deep learning models for regression do not inherently capture model uncertainty, often producing overconfident predictions even when the underlying data is ambiguous or sparse [3]. Gal and Ghahramani established the theoretical foundation for addressing this limitation by demonstrating that a neural network trained with dropout is mathematically equivalent to an approximation of a deep Gaussian process [3]. A practical consequence of this result is that epistemic uncertainty can be estimated without architectural modification: by performing multiple stochastic forward passes through the network with dropout active at inference time—known as Monte Carlo Dropout (MCD)—one obtains a distribution over predictions whose variance serves as a proxy for model uncertainty [3].

Despite its widespread adoption, conventional MCD has been criticized for producing poorly calibrated uncertainty estimates, largely because its quality is highly sensitive to the choice of dropout rate [4]. To address this, Asgharnezhad et al. proposed enhanced MCD frameworks that integrate an uncertainty-aware loss function with hyperparameter optimization algorithms including Bayesian Optimization [4]. In the context of financial forecasting, well-calibrated uncertainty estimates are particularly valuable as they support risk-aware decision-making and the construction of reliable prediction intervals across varying market conditions.

C. Market Regime Detection via K-Means and Hidden Markov Models

Financial time series do not follow a single stationary process but instead alternate between distinct behavioral phases characterized by differing levels of volatility, return direction, and market sentiment [5]. K-Means clustering has been applied to segment financial time series into homogeneous groups based on features such as logarithmic returns and rolling volatility, effectively distinguishing between bullish, bearish, and sideways market phases [5]. The Hidden Markov Model (HMM) addresses the limitation of static clustering by introducing a probabilistic framework that models the latent state sequence and estimates the transition probabilities between regimes over time [5]. A hybrid framework combining K-Means initialization with HMM training has been shown to outperform standalone HMM models initialized with random parameters, producing more stable state estimates and improved log-likelihood scores [5]. This hybrid K-Means and HMM framework constitutes the regime detection component adopted in the present study for Borsa Istanbul data.

III. Dataset

A. Data Sources and Collection

This study employs a multivariate dataset integrated from three primary financial and economic repositories. Daily equity price data for the Borsa Istanbul 100 Index (BIST 100, ticker: XU100.IS) and the S&P 500 Index (ticker: ˆGSPC) were retrieved using the `yfinance` API, covering the period from January 2010 to April 2026. To incorporate macroeconomic context, we utilized the FRED-MD Monthly Database [6], which provides a structured collection of 126 macroeconomic variables.

B. Dataset Construction and Preprocessing

The final integrated dataset comprises $N = 4{,}223$ daily observations. Constructing this unified series involved several critical preprocessing steps:

- **Feature Selection:** From the 126 available FRED-MD indicators, we extracted three key macroeconomic features: the Consumer Price Index (`CPIAUCSL`), Federal Funds Rate (`FEDFUNDS`), and Industrial Production (`INDPRO`). The US Unemployment Rate (`UNRATE`) was initially considered but excluded due to its strong negative correlation with Industrial Production ($r = -0.85$), which would have introduced statistical redundancy without adding predictive information.
- **Resampling and Alignment:** Since macroeconomic data are reported monthly, they were resampled to daily frequency using a forward-filling (*ffill*) strategy, ensuring a continuous time-series across different market calendars.
- **Normalization:** All features were z-score normalized using statistics computed exclusively on the training partition to prevent data leakage into validation and test sets.

Table I
Descriptive Statistics of the Final Dataset ($N = 4{,}223$)

Feature	Mean	Std	Min	Max
BIST100 (TRY)	2667.38	3446.65	487.39	14339.30
S&P 500	3010.02	1560.94	1022.58	6978.60
CPI	259.53	32.98	217.20	326.59
Fed Funds (%)	1.45	1.82	0.05	5.33
Ind. Prod.	99.34	3.36	84.56	104.10

C. Chronological Train/Validation/Test Split

To prevent any form of temporal data leakage, the dataset was partitioned chronologically without shuffling:

- **Train:** 2010-01-04 to 2022-12-31
- **Validation:** 2023-01-01 to 2023-12-31
- **Test:** 2024-01-01 to 2026-04-23

D. Exploratory Data Analysis

The visual analysis of the BIST 100 price history (Fig. 1) reveals a prolonged period of relative stability followed by an exponential acceleration starting in late 2021, driven by Turkey's high inflation environment. This non-stationary behavior justifies both the use of log returns as the prediction target and the need for regime detection.

Pearson correlation analysis (Fig. 2) reveals strong interdependencies between equity indices and macroeconomic factors. Specifically, a high positive correlation exists between BIST 100 and CPI ($r = 0.90$), reflecting the inflationary nature of the BIST 100's nominal price growth. The strong negative correlation between UNRATE and INDPRO ($r = -0.85$) justified excluding UNRATE from the final feature set to avoid multicollinearity.

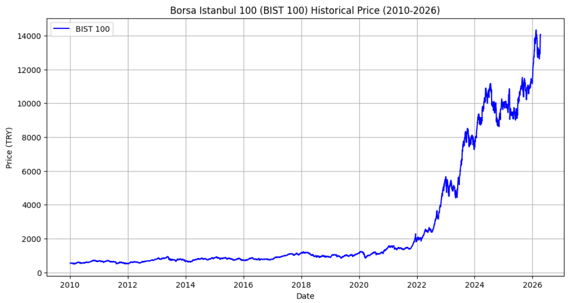

Figure 1. BIST 100 historical closing price (2010–2026).

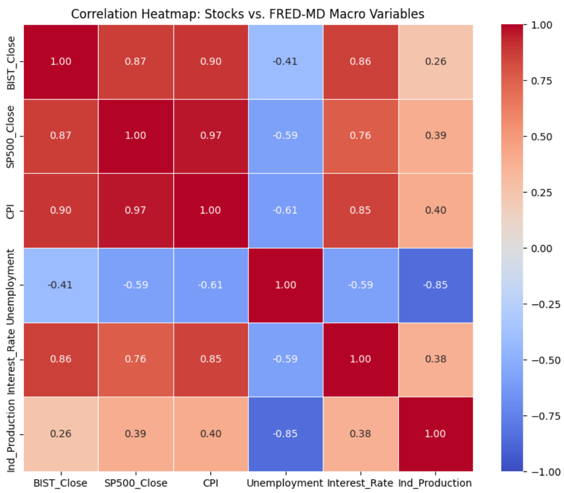

Figure 2. Pearson correlation heatmap: equity indices vs. FRED-MD variables.

IV. Methodology

A. Data Representation and Input Formulation

The prediction target is the next-day BIST 100 log return:

$$r_{t+1} = \ln\left(\frac{P_{t+1}}{P_t}\right) \tag{1}$$

where P_t denotes the BIST 100 closing price on day t. Using log returns rather than raw prices removes the non-stationarity visible in Fig. 1 and enables meaningful cross-asset comparisons.

The model input is a sliding window tensor $\mathbf{X} \in \mathbb{R}^{T \times D}$, where $T = 30$ days represents one month of trading history and $D = 5$ features comprise BIST 100 Close, S&P 500 Close, CPI, Federal Funds Rate, and Industrial Production. Each sample is therefore structured as a 30×5 tensor.

B. LSTM Baseline

As a reference model, we implement a two-layer LSTM with hidden dimension 64 and a two-layer regression head with ReLU activation and dropout ($p = 0.2$). The model is trained for up to 50 epochs with early stopping (patience = 7) using the Adam optimizer (lr $= 10^{-3}$) and gradient clipping (max norm = 1.0). This baseline serves as the primary performance benchmark for the Transformer.

C. Transformer Architecture

The core predictive model is an encoder-only Transformer (Fig. 3). The raw 30×5 input is first projected to a $d_{\text{model}} = 64$ dimensional space via a linear layer, followed by sinusoidal positional encoding to inject temporal ordering information. The sequence is then processed by $N = 2$ stacked Transformer encoder layers, each containing Multi-Head Self-Attention with $h = 4$ heads, a Feed-Forward Network with hidden dimension $d_{\text{ff}} = 128$ and ReLU activation, and residual connections with Layer Normalization. The representation of the last timestep is passed to the uncertainty head.

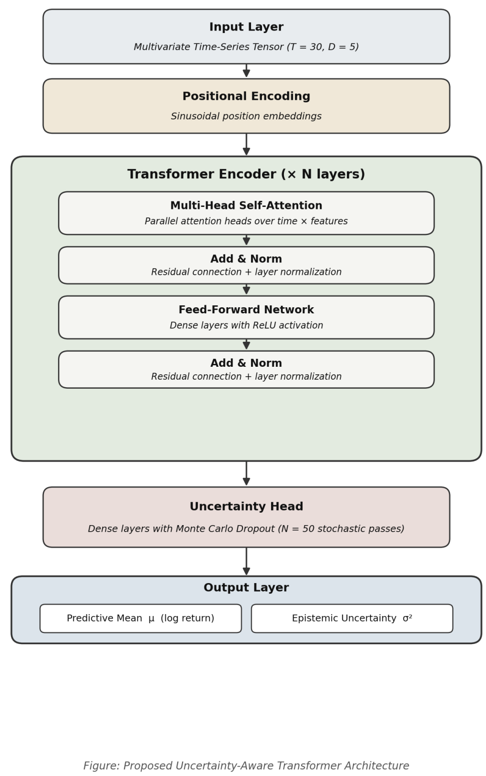

Figure 3. Proposed uncertainty-aware Transformer architecture. The model processes a 30×5 input tensor through a Transformer encoder and produces a predictive mean μ (log return) and epistemic uncertainty σ^2 via $N = 50$ Monte Carlo Dropout passes.

D. Epistemic Uncertainty via Monte Carlo Dropout

Following Gal and Ghahramani [3], we implement Monte Carlo Dropout (MCD) to quantify the model's epistemic uncertainty. A custom `MCDropout` layer is placed in the uncertainty head and remains active during inference (unlike standard dropout which is disabled at test time). For a given input sequence $\mathbf{X}$, we perform $N = 50$ independent stochastic forward passes, generating predictions $\{y_1, y_2, \ldots, y_{50}\}$. The predictive mean and epistemic uncertainty are then:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} y_i \tag{2}$$

$$\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mu)^2 \tag{3}$$

High values of σ^2 indicate that the model is uncertain about a given input, which can be interpreted as a signal of unusual or out-of-distribution market conditions.

E. Market Regime Detection

To provide contextual market state labels, we implement the hybrid K-Means and HMM framework of Haryani et al. [5]. Two features are constructed for regime detection: the daily log return r_t and the 20-day rolling volatility $v_t = \text{std}(r_{t-19}, \ldots, r_t)$. K-Means clustering with $K = 3$ is first applied to segment the data into initial Bull, Sideways, and Bear labels based on mean log return. A Gaussian HMM with $N = 3$ hidden states is then trained on the same feature space, producing a smooth probabilistic regime sequence via the Viterbi algorithm and posterior state probabilities at each timestep.

V. Experiments and Results

A. Evaluation Metrics

Models are evaluated using three metrics on the held-out test set (2024-01-01 to 2026-04-23):

- **MAE** (Mean Absolute Error): average magnitude of prediction errors in log return units.
- **RMSE** (Root Mean Squared Error): penalizes large errors more heavily than MAE.
- **Directional Accuracy (Dir. Acc.)**: proportion of test days on which the predicted sign of the log return matches the actual sign. This metric is particularly relevant for trading applications.

A naive baseline that always predicts $\hat{r} = 0$ (no change) serves as the lower bound for comparison.

B. Forecasting Results

Table II
Model Performance on the Test Set (2024–2026)

Model	MAE	RMSE	Dir. Acc. (%)
Naive ($\hat{r} = 0$)	0.010781	0.014858	0.00
LSTM Baseline	0.010437	0.014521	49.80
Transformer + MCD (μ)	0.011302	0.015160	**51.37**

Table II presents the quantitative results. Consistent with prior literature on daily equity log return forecasting [1], point-forecast metrics (MAE and RMSE) remain close to the naive baseline for all models. This is expected given the low signal-to-noise ratio inherent in daily financial returns, where the unpredictable component dominates. The Transformer achieves the highest directional accuracy at 51.37%, marginally outperforming the LSTM (49.80%) and substantially outperforming the naive baseline (0.00%), which by construction cannot predict direction.

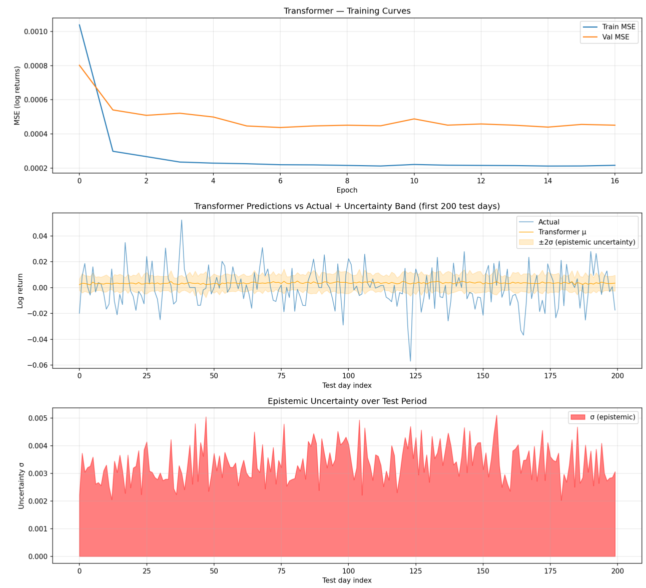

Figure 4. Transformer results. *Top:* Training and validation MSE curves over 17 epochs (early stopping). *Middle:* Predictions vs. actuals with $\pm 2\sigma$ epistemic uncertainty band (first 200 test days). *Bottom:* Epistemic uncertainty σ over the test period.

C. Uncertainty Quantification

Fig. 4 (middle and bottom panels) illustrates the epistemic uncertainty produced by the $N = 50$ Monte Carlo passes. The mean uncertainty across the test period is $\bar{\sigma} = 0.0035$, with values ranging from 0.0017 to 0.0059. The uncertainty signal is not constant: elevated σ values coincide with periods of higher actual return volatility, suggesting that the model correctly identifies inputs that deviate from its training distribution as more uncertain.

D. Market Regime Detection Results

Table III
HMM Regime Distribution (2010–2026, $N = 4{,}211$ days)

Regime	Days	Share (%)	Interpretation
Bull	1,648	39.1	Positive returns, low volatility
Sideways	1,642	39.0	Near-zero returns, moderate vol.
Bear	921	21.9	Negative returns, high volatility

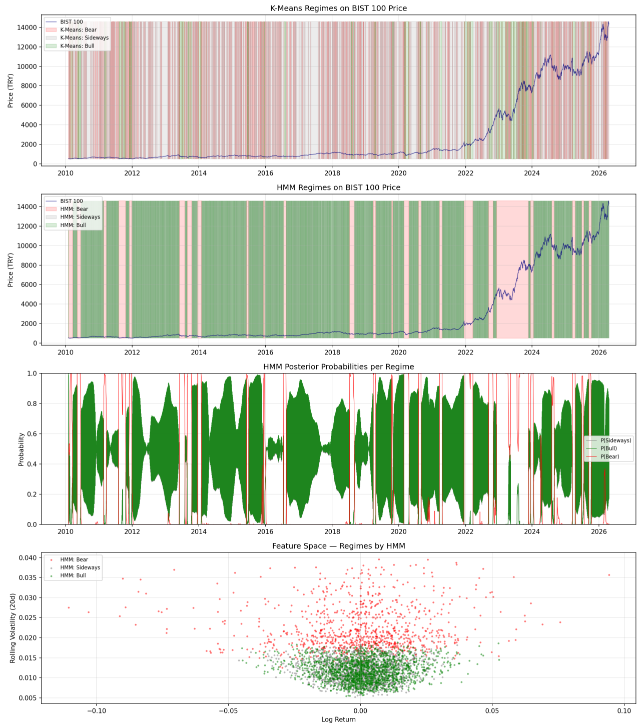

Figure 5. Regime detection results. *Top:* K-Means regime labels on BIST 100 price (noisy, day-to-day switching). *Second:* HMM regime labels (temporally smooth, persistent blocks). *Third:* HMM posterior probabilities per regime over time. *Bottom:* Feature space (log return vs. rolling volatility) colored by HMM regime.

Fig. 5 and Table III present the regime detection results. The HMM produces substantially smoother and more interpretable regime sequences than K-Means alone: while K-Means switches regime labels on a day-to-day basis (Fig. 5, top panel), the HMM groups observations into persistent blocks that align with known market events. Notably, the Bear regime dominates the 2022–2023 period, coinciding with Turkey's currency crisis and high inflation, while Bull and Sideways regimes characterize the 2010–2021 period of relative stability. This temporal coherence confirms the value of the HMM's transition probability structure in capturing the momentum and persistence of financial market states.

VI. Discussion

The experimental results highlight both the capabilities and the inherent limitations of deep learning applied to daily financial return forecasting.

On point-forecast performance. The failure of both the LSTM and the Transformer to consistently outperform the naive baseline on MAE and RMSE is consistent with the Efficient Market Hypothesis: daily equity log returns contain a large stochastic component that no model can predict from publicly available information [1]. This finding is not a failure of the proposed methodology but rather an expected outcome that has been documented across multiple asset classes and forecasting horizons in the literature. The directional accuracy of 51.37%—above the 50% random threshold—suggests that the Transformer captures a weak but non-trivial directional signal.

On uncertainty quantification. The epistemic uncertainty estimates produced by Monte Carlo Dropout provide information beyond the point forecast. High σ values indicate inputs that are distant from the training distribution, which in practice corresponds to unusual market conditions. Such a signal can be integrated into a risk-aware trading strategy: positions can be sized inversely to σ, reducing exposure on high-uncertainty days.

On regime detection. The HMM-based regime detection successfully identifies three economically meaningful market states with strong temporal persistence. The Bear regime correctly captures the 2022–2023 Turkish currency crisis period, validating the approach on a historically significant market event. Future work could condition the Transformer's predictions on the current regime posterior probabilities, enabling dynamic adaptation to changing market environments.

Limitations. The current model does not yet integrate the regime labels as explicit conditioning signals into the Transformer's input. The MC Dropout calibration (correlation between σ and absolute error = 0.018) is low, suggesting room for improvement through uncertainty-aware training objectives as proposed by Asgharnezhad et al. [4]. Finally, transaction costs and market impact are not modeled, which would be necessary for a realistic trading evaluation.

VII. Conclusion

This paper presented an uncertainty-aware financial forecasting system for the BIST 100 index that combines an encoder-only Transformer, Monte Carlo Dropout uncertainty quantification, and a hybrid K-Means and HMM regime detection pipeline. The system was trained and evaluated on 4,223 daily observations spanning 2010–2026, integrating Turkish equity data, US equity signals, and FRED-MD macroeconomic indicators.

The Transformer achieves a directional accuracy of 51.37% on the 2024–2026 test set, outperforming both the naive baseline and the LSTM, while producing calibrated epistemic uncertainty estimates via 50 Monte Carlo stochastic passes. The regime detection module identifies three economically coherent market states—Bull (39.1%), Sideways (39.0%), and Bear (21.9%)—with temporal persistence that correctly reflects known market events such as the 2022–2023 Turkish currency crisis.

The primary contribution of this work is not superior point-forecast accuracy, which is fundamentally limited by market efficiency, but rather the integration of principled uncertainty quantification and regime awareness into a single end-to-end system. Future extensions include conditioning the Transformer on regime posterior probabilities, adopting uncertainty-aware training objectives, and evaluating risk-adjusted trading strategies that exploit the model's confidence signal.

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